Tokyo Medal Ranking and PCA

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Data Description

The 2020 Summer Olympics, held in Tokyo, Japan, from 23 July to 8 August 2021. Overall, there are 93 nations received at least one medal.

```
# load data
library(readr)
data.source<-paste0("https://raw.githubusercontent.com/",
"mcgillstat/regression/main/data/medals_2020.csv")

medals <- read_csv(data.source, col_names = FALSE)
colnames(medals) <- c("Country","Gold","Silver","Bronze")
print(medals,n=93)

## # A tibble: 93 x 4
##    Country            Gold Silver Bronze
##    <chr>             <dbl>  <dbl>  <dbl>
##  1 UnitedStates         39     41     33
##  2 China                38     32     19
##  3 Japan                27     14     17
##  4 GreatBritain         22     20     22
##  5 ROC                  20     28     23
##  6 Australia            17      7     22
##  7 Netherlands          10     12     14
##  8 France               10     12     11
##  9 Germany              10     11     16
## 10 Italy                10     10     20
## 11 Canada                7      7     10
## 12 Brazil                7      6      8
## 13 NewZealand            7      6      7
## 14 Cuba                  7      3      5
## 15 Hungary               6      7      7
## 16 SouthKorea            6      4     10
## 17 Poland                4      5      5
## 18 CzechRepublic         4      4      3
## 19 Kenya                 4      4      2
## 20 Norway                4      2      2
## 21 Jamaica               4      1      4
## 22 Spain                 3      8      6
## 23 Sweden                3      6      0
## 24 Switzerland           3      4      6
## 25 Denmark               3      4      4
## 26 Croatia               3      3      2
## 27 Iran                  3      2      2
## 28 Serbia                3      1      5
## 29 Belgium               3      1      3
```

```
## 30 Bulgaria              3      1      2
## 31 Slovenia              3      1      1
## 32 Uzbekistan            3      0      2
## 33 Georgia               2      5      1
## 34 ChineseTaipei         2      4      6
## 35 Turkey                2      2      9
## 36 Greece                2      1      1
## 37 Uganda                1      1      4
## 38 Ecuador               2      1      0
## 39 Ireland               2      0      2
## 40 Israel                0      2      4
## 41 Qatar                 2      0      1
## 42 Bahamas               2      0      0
## 43 Kosovo                0      0      2
## 44 Ukraine               1      6     12
## 45 Belarus               1      3      3
## 46 Romania               1      3      0
## 47 Venezuela             3      0      4
## 48 India                 1      2      4
## 49 HongKong              1      2      3
## 50 Philippines           1      2      1
## 51 Slovakia              2      1      4
## 52 SouthAfrica           1      2      0
## 53 Austria               1      1      5
## 54 Egypt                 1      1      4
## 55 Indonesia             1      1      3
## 56 Ethiopia              1      1      2
## 57 Portugal              1      2      4
## 58 Tunisia               1      1      0
## 59 Estonia               1      0      1
## 60 Fiji                  0      1      2
## 61 Latvia                0      1      2
## 62 Thailand              0      1      2
## 63 Bermuda               1      0      0
## 64 Morocco               0      0      1
## 65 PuertoRico            0      0      1
## 66 Colombia              0      4      1
## 67 Azerbaijan            0      3      4
## 68 DominicanRepublic     0      3      2
## 69 Armenia               0      2      2
## 70 Kyrgyzstan            0      2      1
## 71 Mongolia              0      1      3
## 72 Argentina             0      1      2
## 73 SanMarino             1      2      3
## 74 Jordan                0      1      1
## 75 Malaysia              1      1      2
## 76 Nigeria               1      1      2
## 77 Bahrain               0      1      0
## 78 Lithuania             1      0      1
```

```
## 79 Namibia               1      0      1
## 80 NorthMacedonia        1      0      1
## 81 SaudiArabia           1      0      1
## 82 Turkmenistan          1      0      1
## 83 Kazakhstan            0      0      8
## 84 Mexico                0      0      4
## 85 Finland               0      0      2
## 86 Botswana              0      0      1
## 87 BurkinaFaso           0      1      1
## 88 Ghana                 0      1      1
## 89 Grenada               0      1      1
## 90 IvoryCoast            0      1      1
## 91 Kuwait                0      1      1
## 92 Moldova               0      1      1
## 93 Syria                 0      1      1
```

Here are summary statistics of the data

```
# PCA applied to medal counts in 2020 Summer Game
# Create a data matrix without 1st column (country name)
X_raw = as.matrix(medals[,-1])
summary(X_raw)
```

```
##       Gold            Silver
##  Min.   : 0.000   Min.   : 0.000
##  1st Qu.: 0.000   1st Qu.: 1.000
##  Median : 1.000   Median : 1.000
##  Mean   : 3.667   Mean   : 3.688
##  3rd Qu.: 3.000   3rd Qu.: 4.000
##  Max.   :39.000   Max.   :41.000
##      Bronze
##  Min.   : 0.000
##  1st Qu.: 1.000
##  Median : 2.000
##  Mean   : 4.634
##  3rd Qu.: 5.000
##  Max.   :33.000
```

Clearly, three variables in data are on the same scale, still we choose to center and scale the data

```
# Standardize the data
X = scale(X_raw, center = T, scale = T)
```

Now we perform the PCA (one dimensional). Perform the eigen-decomposition

$$\boldsymbol{S} = \frac{1}{N}\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{B}\boldsymbol{\Lambda}\boldsymbol{B}^\top$$

to get $\boldsymbol{B} = [\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3]$ and $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$

```r
# Compute the principal components
S = t(X) %*% X / nrow(X)
B = eigen(S)
B

## eigen() decomposition
## $values
## [1] 2.74534630 0.15136108 0.07103456
##
## $vectors
##            [,1]       [,2]         [,3]
## [1,] -0.5819132 -0.3965166  0.710036333
## [2,] -0.5816912 -0.4072185 -0.704136719
## [3,] -0.5683418  0.8227683 -0.006315571
```

This is the matrix $\boldsymbol{B}$ contains principal component vectors

```r
B$vectors

##            [,1]       [,2]         [,3]
## [1,] -0.5819132 -0.3965166  0.710036333
## [2,] -0.5816912 -0.4072185 -0.704136719
## [3,] -0.5683418  0.8227683 -0.006315571
```

The first principle component $\boldsymbol{b}_1$ is the first column of the matrix $\boldsymbol{B}$.

```r
# First principal component (PC1) is the 1st column of matrix B
b1 = B$vectors[,1]
```

Now, for any country n, we can project the vector of medal counts $\boldsymbol{x}_n$

$$\boldsymbol{x}_n = [\text{gold}_n, \text{silver}_n, \text{bronze}_n]^\top$$

onto $\boldsymbol{b}_1$ to obtain the coordinates of the project observations

$$z_{1n} = \boldsymbol{b}_1^\top \boldsymbol{x}_n = -0.581 \times \text{gold}_n - 0.581 \times \text{silver}_n - 0.568 \times \text{bronze}_n$$

for $n = 1, 2, 3$ (US, China, Japan). Notice that the corresponding weights are quite similar for gold, silver and bronze medals, suggesting their impacts on the data variation are almost equal.

```r
# coordinates of PC1 for all countries
z1 = X %*% b1
dim(z1)

## [1] 93  1

# The coordinate of US for the first component
z1[1]

## [1] -8.879672
```

```
X[1,] %*% b1

##           [,1]
## [1,] -8.879672

# The coordinate of China for the first component
z1[2]

## [1] -6.692825

# The coordinate of Japan for the first component
z1[3]

## [1] -4.00174
```

Let examine the proportion of variance explained by the first PC:

```
# The amount of variance explained by PC1
B$values[1]/(B$values[1]+B$values[2]+B$values[3])

## [1] 0.9250623
```

We obtain all the coordinates $z_{11}, z_{12}, \ldots, z_{1N}$ for $N = 93$ countries, using which as scores, we rank the countries.

```
# Ranking by coordinates on first PC
result1 = rank(z1, ties.method='min')
result1

##  [1]  1  2  5  4  3  6  9 10  8  7 11 12 15 18 14
## [16] 13 19 24 25 32 27 17 28 21 23 31 41 26 40 47
## [31] 52 51 30 22 20 62 43 70 60 42 69 88 71 16 37
## [46] 61 39 35 45 59 38 68 34 43 49 56 35 87 81 63
## [61] 63 63 93 89 89 50 33 48 55 67 54 63 45 73 56
## [76] 56 92 81 81 81 81 81 29 53 71 89 73 73 73 73
## [91] 73 73 73
```

Note that here we used `rank()` function to assign a rank to each element in a vector. To handle ties with `rank()` function, you can use the `ties.method` argument to specify how we should handle ties. In the code, we set `ties.method='min'`, this will assign every tied element to the lowest rank (elements ranked in the 3rd and 4th position would both receive a rank of 3). For details, see `https://www.statology.org/sort-order-rank-in-r/`.

Let's see how diffferent the results are, based on different ranking methods

```
# Ranking by total count
score2 = as.vector(-X_raw%*%matrix(c(1,1,1),nrow=3))
result2 = rank(score2, ties.method='min')

# Ranking by gold count
```

```
score3 = -X_raw%*%matrix(c(1,0,0),nrow=3)
result3 = rank(score3, ties.method='min')

# ranking difference PC1 vs total
result1 - result2

##  [1]  0  0  0  0  0  0  0  0  0  0  0  0  2  0  1
## [16]  0  0  1  0  3  1  0  2  1  0  2  8  0  7  5
## [31]  4  3  1  0  0  9  1  7  7  0  6 17  0  0  4
## [46]  8  6  2  3  6  5  5  1  1  1  3  2 16 10  0
## [61]  0  0  4  0  0  2  0  0  2  4  1  0  3  2  3
## [76]  3  3 10 10 10 10 10  0  0  0  0  2  2  2  2
## [91]  2  2  2

# ranking difference PC1 vs gold
result1 - result3

##  [1]   0   0   2   0  -2   0   2   3   1   0   0
## [12]   1   4   7  -1  -2   2   7   8  15  10  -5
## [23]   6  -1   1   9  19   4  18  25  30  29  -4
## [34] -12 -14  28   0  36  26 -25  35  54   4 -27
## [45]  -6  18  17  -8   2  16   4  25  -9   0   6
## [56]  13  -8  44  38  -4  -4  -4  50  22  22 -17
## [67] -34 -19 -12   0 -13  -4   2   6  13  13  25
## [78]  38  38  38  38  38 -38 -14   4  22   6   6
## [89]   6   6   6   6   6
```

We can see that the ranking result from the PCA is quite similar to that from the total count, but quite different to that based on the gold medal count.

Here is the summary of three different ranking results

```
# Add three ranking results into the data
dat1 = data.frame(X_raw)
rownames(dat1) <- medals$Country
dat1$PC1 = result1
dat1$Tot = result2
dat1$By_G = result3
dat1

##                  Gold Silver Bronze PC1 Tot By_G
## UnitedStates       39     41     33   1   1    1
## China              38     32     19   2   2    2
## Japan              27     14     17   5   5    3
## GreatBritain       22     20     22   4   4    4
## ROC                20     28     23   3   3    5
## Australia          17      7     22   6   6    6
## Netherlands        10     12     14   9   9    7
## France             10     12     11  10  10    7
## Germany            10     11     16   8   8    7
## Italy              10     10     20   7   7    7
```

```
## Canada              7      7     10  11  11   11
## Brazil              7      6      8  12  12   11
## NewZealand          7      6      7  15  13   11
## Cuba                7      3      5  18  18   11
## Hungary             6      7      7  14  13   15
## SouthKorea          6      4     10  13  13   15
## Poland              4      5      5  19  19   17
## CzechRepublic       4      4      3  24  23   17
## Kenya               4      4      2  25  25   17
## Norway              4      2      2  32  29   17
## Jamaica             4      1      4  27  26   17
## Spain               3      8      6  17  17   22
## Sweden              3      6      0  28  26   22
## Switzerland         3      4      6  21  20   22
## Denmark             3      4      4  23  23   22
## Croatia             3      3      2  31  29   22
## Iran                3      2      2  41  33   22
## Serbia              3      1      5  26  26   22
## Belgium             3      1      3  40  33   22
## Bulgaria            3      1      2  47  42   22
## Slovenia            3      1      1  52  48   22
## Uzbekistan          3      0      2  51  48   22
## Georgia             2      5      1  30  29   34
## ChineseTaipei       2      4      6  22  22   34
## Turkey              2      2      9  20  20   34
## Greece              2      1      1  62  53   34
## Uganda              1      1      4  43  42   43
## Ecuador             2      1      0  70  63   34
## Ireland             2      0      2  60  53   34
## Israel              0      2      4  42  42   67
## Qatar               2      0      1  69  63   34
## Bahamas             2      0      0  88  71   34
## Kosovo              0      0      2  71  71   67
## Ukraine             1      6     12  16  16   43
## Belarus             1      3      3  37  33   43
## Romania             1      3      0  61  53   43
## Venezuela           3      0      4  39  33   22
## India               1      2      4  35  33   43
## HongKong            1      2      3  45  42   43
## Philippines         1      2      1  59  53   43
## Slovakia            2      1      4  38  33   34
## SouthAfrica         1      2      0  68  63   43
## Austria             1      1      5  34  33   43
## Egypt               1      1      4  43  42   43
## Indonesia           1      1      3  49  48   43
## Ethiopia            1      1      2  56  53   43
## Portugal            1      2      4  35  33   43
## Tunisia             1      1      0  87  71   43
## Estonia             1      0      1  81  71   43
```

```
## Fiji                 0      1      2  63  63   67
## Latvia               0      1      2  63  63   67
## Thailand             0      1      2  63  63   67
## Bermuda              1      0      0  93  89   43
## Morocco              0      0      1  89  89   67
## PuertoRico           0      0      1  89  89   67
## Colombia             0      4      1  50  48   67
## Azerbaijan           0      3      4  33  33   67
## DominicanRepublic    0      3      2  48  48   67
## Armenia              0      2      2  55  53   67
## Kyrgyzstan           0      2      1  67  63   67
## Mongolia             0      1      3  54  53   67
## Argentina            0      1      2  63  63   67
## SanMarino            1      2      3  45  42   43
## Jordan               0      1      1  73  71   67
## Malaysia             1      1      2  56  53   43
## Nigeria              1      1      2  56  53   43
## Bahrain              0      1      0  92  89   67
## Lithuania            1      0      1  81  71   43
## Namibia              1      0      1  81  71   43
## NorthMacedonia       1      0      1  81  71   43
## SaudiArabia          1      0      1  81  71   43
## Turkmenistan         1      0      1  81  71   43
## Kazakhstan           0      0      8  29  29   67
## Mexico               0      0      4  53  53   67
## Finland              0      0      2  71  71   67
## Botswana             0      0      1  89  89   67
## BurkinaFaso          0      1      1  73  71   67
## Ghana                0      1      1  73  71   67
## Grenada              0      1      1  73  71   67
## IvoryCoast           0      1      1  73  71   67
## Kuwait               0      1      1  73  71   67
## Moldova              0      1      1  73  71   67
## Syria                0      1      1  73  71   67
```

All analysis above can be done through a simple function in R, i.e. `prcomp`

```
# Alternatively, you can use function prcomp to perform PCA analysis
pca_result = prcomp(~ Gold + Silver + Bronze, data = dat1,  center = T, scale= T)

# coordinates of PC1 for all countries
pca_result$x[,1]

##      UnitedStates             China
##       -8.87967167       -6.69282455
##             Japan      GreatBritain
##       -4.00174004       -4.58505818
##               ROC         Australia
##       -5.22035889       -3.02053352
##       Netherlands            France
```

```
##       -2.13520920       -1.85496936
##           Germany             Italy
##       -2.23357384       -2.51876505
##            Canada            Brazil
##       -1.07053464       -0.79524617
##        NewZealand              Cuba
##       -0.70183289       -0.24962058
##           Hungary        SouthKorea
##       -0.70739085       -0.72224495
##            Poland    CzechRepublic
##       -0.17783256        0.09745592
##             Kenya            Norway
##        0.19086920        0.36779303
##           Jamaica             Spain
##        0.26942839       -0.45372763
##            Sweden       Switzerland
##        0.28367588       -0.09987997
##           Denmark           Croatia
##        0.08694659        0.36223507
##              Iran            Serbia
##        0.45069698        0.25891906
##           Belgium          Bulgaria
##        0.44574562        0.53915890
##          Slovenia        Uzbekistan
##        0.63257218        0.62762082
##           Georgia     ChineseTaipei
##        0.36162847       -0.01697602
##            Turkey            Greece
##       -0.12029203        0.71547613
##            Uganda           Ecuador
##        0.51814024        0.80888941
##           Ireland            Israel
##        0.71052477        0.51258228
##             Qatar           Bahamas
##        0.80393805        0.89735133
##            Kosovo           Ukraine
##        0.87633267       -0.67147558
##           Belarus           Romania
##        0.43462969        0.71486953
##         Venezuela             India
##        0.44079425        0.42967832
##          HongKong       Philippines
##        0.52309161        0.70991817
##          Slovakia      SouthAfrica
##        0.43523629        0.80333145
##           Austria             Egypt
##        0.42472696        0.51814024
##         Indonesia          Ethiopia
##        0.61155352        0.70496680
```

```
##          Portugal           Tunisia
##        0.42967832        0.89179336
##           Estonia              Fiji
##        0.88684200        0.78787075
##            Latvia          Thailand
##        0.78787075        0.78787075
##           Bermuda           Morocco
##        0.98025528        0.96974595
##        PuertoRico          Colombia
##        0.96974595        0.61589829
##        Azerbaijan DominicanRepublic
##        0.42412036        0.61094692
##           Armenia        Kyrgyzstan
##        0.69940884        0.79282212
##          Mongolia         Argentina
##        0.69445747        0.78787075
##         SanMarino            Jordan
##        0.52309161        0.88128403
##          Malaysia           Nigeria
##        0.70496680        0.70496680
##           Bahrain         Lithuania
##        0.97469731        0.88684200
##           Namibia    NorthMacedonia
##        0.88684200        0.88684200
##       SaudiArabia      Turkmenistan
##        0.88684200        0.88684200
##        Kazakhstan            Mexico
##        0.31585299        0.68950611
##           Finland          Botswana
##        0.87633267        0.96974595
##       BurkinaFaso             Ghana
##        0.88128403        0.88128403
##           Grenada        IvoryCoast
##        0.88128403        0.88128403
##            Kuwait           Moldova
##        0.88128403        0.88128403
##             Syria
##        0.88128403

# First principal component (PC1)
pca_result$rotation[,1]

##       Gold     Silver     Bronze
## -0.5819132 -0.5816912 -0.5683418

summary(pca_result)

## Importance of components:
##                           PC1    PC2     PC3
## Standard deviation     1.6659 0.3912 0.26797
```

```
## Proportion of Variance 0.9251 0.0510 0.02394
## Cumulative Proportion  0.9251 0.9761 1.00000
```

```
plot(pca_result)
```

